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Applications Note

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Sequence analysis

PrimerDesign-M: a multiple-alignment based multiple-primer design tool for walking across variable genomes

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Abstract

Summary: Analyses of entire viral genomes or mtDNA requires comprehensive design of many primers across their genomes. Furthermore, simultaneous optimization of several DNA primer design criteria may improve overall experimental efficiency and downstream bioinformatic processing. To achieve these goals, we developed PrimerDesign-M. It includes several options for multiple-primer design, allowing researchers to efficiently design walking primers that cover long DNA targets, such as entire HIV-1 genomes, and that optimizes primers simultaneously informed by genetic diversity in multiple alignments and experimental design constraints given by the user. PrimerDesign-M can also design primers that include DNA barcodes and minimize primer dimerization. PrimerDesign-M finds optimal primers for highly variable DNA targets and facilitates design flexibility by suggesting alternative designs to adapt to experimental conditions.
Availability and implementation: PrimerDesign-M is available as a webtool at http://www.hiv.lanl.gov/content/sequence/PRIMER_DESIGN/primer_design.html
Contact: tkl@lanl.gov or seq-info@lanl.gov.

1 Introduction

Primer design is complicated by many factors, including desired primer specificity or promiscuity, restrictions in polymerase chain reaction (PCR) or sequencing lengths, matching of melting temperatures, heterogeneous evolutionary rates across genomes, avoiding primer–primer dimerization and integrating bio-barcodes. Typically, the process of finding the optimal placements for primers is a serial process in which each of the design criteria are optimized one at a time. This is unfortunate as one design step may influence another. Thus, designing primers while optimizing all design criteria simultaneously is desirable. Because no existing software was able to do that, we recently developed a tool that could include all these factors, given a multiple-alignment and user input on desired constraints, to design primer pairs for single-fragment analyses (Brodin *et al.*, 2013). However, with the increased abilities of next-generation sequencing (NGS) a need to design optimal multiple-fragment analyses, such as tiled fragments that cover entire small genomes, has rapidly emerged.

Many previous primer design software have focused on special purposes such as multiplexing, degenerate sites, discriminate amplification, nested PCR, single-nucleotide polymorphism (SNP) protocols, and hybridization analyses such as micro-arrays and *in situ* hybridization, and DNA sequencing (Fredslund *et al.*, 2006; Giegerich *et al.*, 1996; Kalendar *et al.*, 2011; Rouillard *et al.*, 2003; Rychlik 2007; Untergasser *et al.*, 2012; Vallone and Butler, 2004; Weckx *et al.*, 2005). More generic software, such as Primer3 (Untergasser *et al.*, 2012), base their design on a single sequence, which inadvertently could place primers in more variable regions, leading to annealing failure with some templates or selective amplification in diverse populations.

Motivated by the large genetic diversity found in Human immunodeficiency virus (HIV) and Simian immunodeficiency virus (SIV), and the need for a generic design tool that could dynamically optimize primers across their genomes, we developed PrimerDesign-M. It can be used to design single- or multiple-fragment primers to any aligned set of DNA sequences, regardless of diversity level and organism.

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2 Overview of PrimerDesign-m

2.1 Input

The input consists of a multiple-alignment and user-specified primer design criteria, uploaded and entered into a web interface. The alignment can either be user provided or directly imported from the premade HIV or SIV gene or genome alignments available at the Los Alamos HIV sequence database. To avoid placing primers in gappy regions, gaps may be removed at a user-provided cut-off.

The user selects a *region of interest* (ROI), which defines the part of the alignment that must be covered by the primers. Note that the alignment must be longer than the ROI so that flanking primers can be designed. Next, the user chooses whether a single pair or multiple pairs of primers should cover the ROI. Adaptors can be added, and expected experimental read lengths are defined. If multiple-primer design is selected, five different design options become available: *Max* for maximum overlap between fragments, useful when one wants to ensure contig coverage even when some fragments may fail, *Mid* for a 'tiled' fragment design, *Min* for minimum overlap, *Zero* for no overlap, i.e. 3′-primers are reused as 5′-primers in next fragment and *Flex* for moving across the ROI placing forward primers in positions with as low as possible entropy scores anywhere within the previous fragment.

DNA barcode tags can be added, optimizing either number of tags, tag length or tag edit distance. Additional design options include primer lengths, primer detection limit (design inclusion of degenerated primer sites to cover genetic diversity), complexity limit (maximum number of primer species caused by degenerated sites), maximum difference in T_m between primers, G/C clamps and dimerization risk [dimer window size to be considered, and maximum ratio of matched nucleotides within the dimer window; the default dimerization values are based on experimental results (Desmarais *et al.*, 2012)]. The dimerization test includes all possible homo- and hetero-dimer formations, including hairpin structures. T_m's are calculated using the empirical nearest neighbour model (SantaLucia *et al.*, 1996). For designs that may take more than a minute to design, the user inputs a mail-back email address.

All input options have a link that explains parameter definitions and value ranges. As usual in HIV database tools, we also provide example input to test run the tool.

2.2 Output

The output consists of a summary of design criteria, one or several design graphic representations (depending on if single- or multiple-fragment design was chosen), tables with primer designs, and download options for graphics, tables, gap-stripped alignment and dimerization test logs. We provide five alternative primer pairs per fragment to allow troubleshooting and potential alternatives, should one set experimentally fail. The list of primer pairs is ordered by least entropy among primers that fulfill all design criteria. This should make it easy to continue with an otherwise successful design. Graphics are available in png, pdf and jpg formats, and tables can be downloaded in txt format to be read by spreadsheet or database programs. For multiple- fragment design, a summary graphic shows the overall design including the ROI and the locations of the first alternative set of primers per fragment, curves for entropy, complexity and T_m (Fig. 1).

If mail-back was used, the link to the results stays active for 4 days. If the design fails at some point during multiple-fragment optimization, the output will still show successful fragment design up until the failed fragment. This should allow the user to modify the design criteria that can cover the whole ROI or define a new ROI for the remaining sequence.

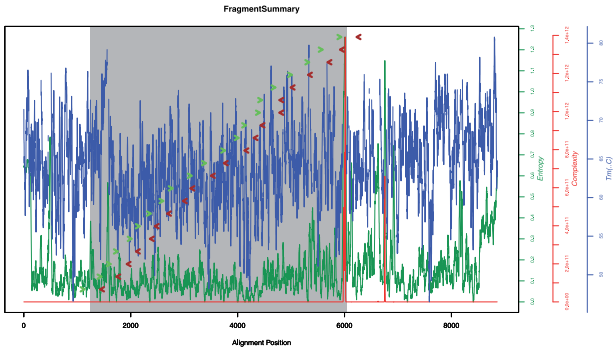

Fig. 1. Multiple fragment design of walking primers to analyze a large part of the genome of HIV-1 subtype G. The *x*-axis shows alignment position according to the original input multiple alignment, and *y*-axes show entropy (green), complexity (red, log scale), and *Tm* (blue). The grey region is the user-specified ROI. Primer pairs for each designed fragment are displayed as green (forward) and red (reverse) arrowheads

2.3 Example of use

Figure 1 shows a potential walking-primer design to amplify/sequence HIV-1 subtype G. Here, we used the HIV database 'web alignment' of 2013, and a ROI in positions 2000–7000 with gaps removed if more frequent than 50% in any alignment column. We used the *Flex* fragment overlap option as we wanted to make sure primers sit in regions with as low diversity as possible given other constraints, especially read length; we wanted to use custom sequencing adaptors ('My adaptor' input was M13 forward adaptor CGCCAGGGTTTTCCCAGTCACGAC and reverse adaptor AGCGGATAACAATTTCACACAGGA); an expected read length 300–400 nt; no barcode tags; and default settings otherwise (minimum primer length = 20, maximum primer length = 25, detection limit = 5%, complexity limit = 32, maximum difference in $T_m = 5°C$, dimer window size = 10, dimer max ratio = 0.9, G/C clamp on forward and reverse primers).

The design resulted in 21 fragments 300–400 nt long covering the ROI, which was modified to positions 1242–6032 when gaps were removed. The resulting gapstripped alignment could be downloaded. For each of the 21 fragments a table with five alternative pairs was provided. It took about 2 min for PrimerDesign-M to complete this design.

3 Conclusion

PrimerDesign-M can design primers for both single fragment analyses as well as multiple-fragment walking-primer analyses of highly variable genomes, and the designed primers have been experimentally confirmed to work (Brodin *et al.*, 2013; Fischer *et al.*, 2010). PrimerDesign-M should be useful to researchers in all fields of biology who need to design DNA primers. To make access easy, guarantee that the latest version of the tool is used and make the program independent of computer platform, the software is available as a web tool at the LANL HIV database.

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Conflict of Interest: none declared.

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References

Brodin,J. *et al*. (2013) A multiple-alignment based primer design algorithm for genetically highly variable DNA targets. *BMC Bioinformatics*, **14**, 255.

Desmarais,S.M. *et al*. (2012) Quantitative experimental determination of primer-dimer formation risk by free-solution conjugate electrophoresis. *Electrophoresis*, **33**, 483–491.

Fischer,W. *et al*. (2010) Transmission of single HIV-1 genomes and dynamics of early immune escape revealed by ultra-deep sequencing. *PLoS One*, **5**, e12303.

Fredslund,J. *et al*. (2006) A general pipeline for the development of anchor markers for comparative genomics in plants. *BMC Genomics*, **7**, 207.

Giegerich,R. *et al*. (1996) GeneFisher-software support for the detection of postulated genes. *Proc. Int. Conf. Intell. Syst. Mol. Biol.*, **4**, 68–77.

Kalendar,R. *et al*. (2011) Java web tools for PCR, in silico PCR, and oligonucleotide assembly and analysis. *Genomics*, **98**, 137–144.

Rouillard,J.M. *et al*. (2003) OligoArray 2.0: design of oligonucleotide probes for DNA microarrays using a thermodynamic approach. *Nucleic Acids Res.*, **31**, 3057–3062.

Rychlik,W. (2007) OLIGO 7 primer analysis software. *Methods Mol. Biol.*, **402**, 35–60.

SantaLucia,J. Jr *et al*. (1996) Improved nearest-neighbor parameters for predicting DNA duplex stability. *Biochemistry*, **35**, 3555–3562.

Untergasser,A. *et al*. (2012) Primer3–new capabilities and interfaces. *Nucleic Acids Res.*, **40**, e115.

Vallone,P.M. and Butler,J.M. (2004) AutoDimer: a screening tool for primer-dimer and hairpin structures. *Biotechniques*, **37**, 226–231.

Weckx,S. *et al*. (2005) SNPbox: a modular software package for large-scale primer design. *Bioinformatics*, **21**, 385–387.

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